Pairs Trading via Ornstein–Uhlenbeck Mean Reversion

A Continuous-Time Stochastic Calculus Approach

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Stochastic Calculus for Finance – Course Project

1. Motivation and Theoretical Framework

Individual equity prices are typically modelled as geometric Brownian motions and are therefore non-stationary. Yet for certain pairs of stocks in the same industry, a linear combination of their log-prices is empirically *stationary*. Such pairs are *cointegrated* (Engle & Granger, 1987); the stationary combination (the *spread*) is a natural candidate for a mean-reverting SDE. We model the spread as an Ornstein–Uhlenbeck (OU) process, calibrate it to training data, and trade a rule that enters when the spread is far from its mean and exits when it reverts.

The OU SDE. On $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ supporting a standard Brownian motion $W(t)$ (Shreve [1], Ch. 3), the OU process $X(t)$ satisfies

$$\mathrm{d}X(t) = \theta\big(\mu - X(t)\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W(t), \qquad X(0) = x_0, \tag{1}$$

with $\theta > 0$ (mean-reversion speed), $\mu \in \mathbb{R}$ (long-run mean), $\sigma > 0$ (volatility). Equation (1) is the Vasicek SDE of Shreve Example 4.4.10 under $\theta = \beta, \mu = a/\beta$.

Closed-form solution via the Itô–Doeblin formula. Let $Y(t) = e^{\theta t}X(t)$. Applying the Itô–Doeblin formula (Shreve Thm. 4.4.1) to $f(t,x) = e^{\theta t}x$ ($f_t = \theta e^{\theta t}x$, $f_x = e^{\theta t}$, $f_{xx} = 0$) and substituting (1), the $X(t)\,\mathrm{d}t$ terms cancel to leave $\mathrm{d}Y(t) = \theta\mu e^{\theta t}\mathrm{d}t + \sigma e^{\theta t}\mathrm{d}W(t)$. Integrating and dividing by $e^{\theta t}$ gives

$$X(t) = x_0 e^{-\theta t} + \mu\big(1 - e^{-\theta t}\big) + \sigma \int_0^t e^{-\theta(t-s)}\,\mathrm{d}W(s). \tag{2}$$

The first two terms describe deterministic exponential decay of the initial deviation toward μ; the third is an Itô integral (Shreve §4.3) against the exponential kernel. By the Itô isometry $\mathrm{Var}[X(t)] = (\sigma^2/2\theta)(1 - e^{-2\theta t})$, so as $t \to \infty$, $X(t) \Rightarrow \mathcal{N}(\mu, \sigma^2/(2\theta))$. The half-life of mean reversion is $\tau = \ln 2/\theta$, and the stationary standard deviation $\sigma_s = \sigma/\sqrt{2\theta}$ quantifies what "far from the mean" means.

From GBM prices to a stationary spread. Each $\log S_i(t)$ has unbounded variance. But if two log-prices share a common stochastic trend, a cointegrating vector $(1, -\beta)$ exists such that $Z(t) = \log S_A(t) - \alpha - \beta \log S_B(t)$ is stationary, and $Z(t)$ can be modelled by (1).

2. Methodology

Data. Daily adjusted-close prices for eight candidate US/Indian pairs (Yahoo Finance, 2021-01-01 to 2024-12-31). First 75% of days used for estimation (*training*), last 25% reserved for *out-of-sample* backtest. No test data informs the screen, the fit, or the thresholds.

Step 1 – Cointegration screen. For each pair we run the Engle–Granger two-step procedure: OLS of $\log S_A$ on $\log S_B$ yields (α, β); an Augmented Dickey–Fuller test is then run on the residuals. Accept at the 5% level if ADF < -3.34 (MacKinnon).

Step 2 – OU calibration. The exact discretisation of (1) is AR(1): $X_{t+\Delta t} = aX_t + b + \eta_t$ with $a = e^{-\theta\Delta t}$, $b = \mu(1-a)$. OLS of $X_{t+\Delta t}$ on X_t with $\Delta t = 1/252$ gives $\theta = -\ln(a)/\Delta t$, $\mu = b/(1-a)$, $\sigma = \sqrt{2\theta\,\mathrm{Var}(\eta)/(1-a^2)}$.

Step 3 – Trading rule. Let $Z_t = (X_t - \mu)/\sigma_s$. **Enter** when $|Z_t| > 2$ (short spread if $Z_t > 2$, long if $Z_t < -2$); **take profit** at $|Z_t| < 0.5$; **stop-loss** at $|Z_t| > 3.5$; **time-stop** after 2τ. Re-entry is blocked until $|Z_t|$ re-enters ± 2 (prevents whipsaw). Thresholds are set *a priori* from the stationary distribution of Z and never tuned.

Costs. Leg A at unit notional, leg B at β; 10 bp per leg per trade, so round-trip cost $= 2(1 + |\beta|) \times 10$ bp.

3. Empirical Results

Pair screening. Table 1 reports the eight pairs. Two pass the 5% cointegration test: **MA/V** (ADF $= -3.77$, half-life 19.5 d) and **RELIANCE.NS/ONGC.NS** (ADF $= -3.39$, half-life 25.7 d). The six rejected pairs confirm that sectoral similarity does not imply cointegration.

Table 1: Engle–Granger test and OU parameters (training window, 756 days). Bold rows pass at 5%.

Pair	β	ADF	Coint.	θ /yr	σ_s	Half-life (d)
KO/PEP	0.738	−2.46	No	3.72	0.036	47.0
XOM/CVX	1.243	−1.76	No	1.92	0.082	91.1
MA/V	**0.952**	**-3.77**	**Yes**	**8.96**	**0.029**	**19.5**
GLD/SLV	0.231	−1.30	No	1.57	0.056	111.6
HD/LOW	0.839	−2.31	No	3.85	0.046	45.3
HDFCBANK.NS/ICICIBANK.NS	0.248	−3.15	No	6.31	0.055	27.7
TCS.NS/INFY.NS	0.467	−2.02	No	3.48	0.057	50.1
RELIANCE.NS/ONGC.NS	**0.336**	**-3.39**	**Yes**	**6.79**	**0.061**	**25.7**

Backtest (out-of-sample, 252 days). Table 2 gives the aggregate results. Three trades: MA/V produced one clean reversion win and one stop-loss; RELIANCE/ONGC produced one small win. Combined success ratio $2/3 = 66.7\%$, net P&L +2.07% after costs.

Table 2: Out-of-sample backtest results (2023-11-27 to 2024-12-31).

Pair	# trades	wins	success	avg %	total %	Sharpe	max DD%	avg hold (d)
MA/V	2	1	0.500	1.00	2.01	0.53	−2.67	33
RELIANCE.NS/ONGC.NS	1	1	1.000	0.07	0.07	–	0.00	26
Combined	**3**	**2**	**0.667**	**0.69**	**2.07**	–	**-2.67**	**31**

Illustration – MA/V. Figure 1 (left) shows the test-set spread and z-score with trade markers. Trade 1 (Mar–May 2024, green) is a textbook reversion: entry at $z \approx +2.2$, the spread relaxes to μ over $\sim$30 days, exit at $+0.5\sigma_s$ for +4.7%. Trade 2 (Aug–Sep 2024, red) is the opposite: after entry at $z \approx +2.5$, the spread drifts to $+4\sigma_s$, triggering the $3.5\sigma_s$ stop at −2.7%. The top panel shows the spread breaking through the stop band before eventually reverting in November, consistent with a temporary cointegration breakdown – precisely the scenario the stop-loss is designed to contain. Figure 1 (right) plots the cumulative P&L curves.

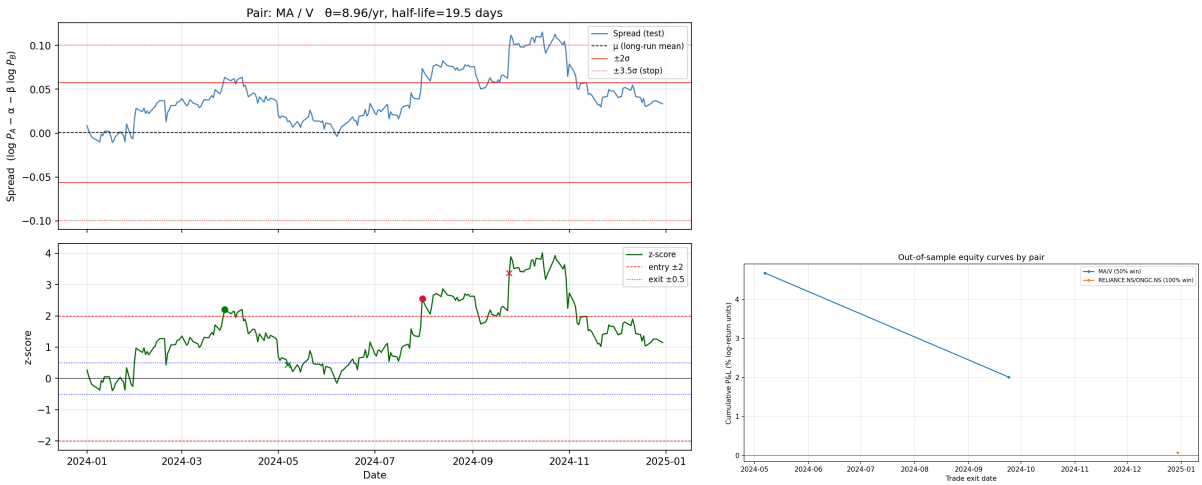

Figure 1: *Left:* MA/V out-of-sample spread (top) and z-score (bottom). Dots = entries, crosses = exits. *Right:* cumulative out-of-sample P&L by pair, net of costs.

4. Discussion and Limitations

What worked. The cointegration filter has teeth: six of eight sector-matched candidates were rejected, and both retained pairs were profitable out-of-sample. The MA/V winner matches the OU prediction almost exactly – entry just above $+2\sigma_s$, monotone drift to μ, take-profit exit – showing that when the stationarity assumption holds, the edge is real.

What failed. The MA/V stop-loss exposes the main model risk: OU parameters estimated in training need not remain valid out-of-sample. Cointegration can drift or break under regime shifts. The $3.5\sigma_s$ stop contained the loss at −2.7%; without it, the position would have stayed open for roughly three additional months in drawdown.

Key limitations. *Sample size:* three trades across two pairs is too few for precise claims; 66.7% is directional evidence, not a point estimate. *Survivorship bias:* candidates were chosen from well-known sector analogues, biasing the screen toward relationships that remain recognisable today. *Static β:* a Kalman-filtered hedge ratio that updates online would track drift and likely

reduce stop-out frequency. *Cost sensitivity:* at 20 bp/leg (more representative of Indian equities with STT) roughly half the MA/V profit would be eroded and the RELIANCE trade would turn negative.

Conclusion. The OU model, derived directly from the Itô–Doeblin formula applied to the cointegration residual, provides a coherent bridge from the continuous-time machinery in Shreve Chs. 3–4 to a concrete trading strategy. The out-of-sample evidence is modestly positive (+2.07% net in one year, 2/3 hit rate) and, more importantly, behaves as theory predicts: one clean reversion, one stop-loss where the stationarity assumption temporarily failed. Together these two outcomes form a compact demonstration of both the power and the limits of continuous-time mean-reversion modelling in practice.

References

[1] Shreve, S. E. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models.* Springer.

[2] Engle, R. F. & Granger, C. W. J. (1987). Co-integration and error correction. *Econometrica* 55(2), 251–276.

[3] Gatev, E., Goetzmann, W. N. & Rouwenhorst, K. G. (2006). Pairs trading: performance of a relative-value arbitrage rule. *RFS* 19(3), 797–827.

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